METRIC

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February 9, 1993
SUPERSEDING
MIL-B-45326C(AT)
2 June 1989

COMMERCIAL ITEM DESCRIPTION

BOOT, DUST AND MOISTURE SEAL: HYDRAULIC BRAKE CYLINDER, SYNTHETIC RUBBER

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-B-45326C(AT) which is canceled.

1.0 Abstract. This CID covers brake boots used in covering the open ends of wheel and master cylinders in hydraulic brake systems of automobiles, trucks, and other wheeled vehicles employing hydraulic brake fluid of the nonpetroleum base type. Brake boots covered by this specification are intended for use as protective devices against dirt and moisture.

2.0 Salient characteristics.

2.1 Materials. Materials shall be a synthetic, non-oil resistant rubber compound conforming to ASTM D2000 M4AA 510 A13B13 or ASTM D2000 M4AA 610 A13B13. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.4).

2.2 Design and construction. Boots shall be of molded construction and shall conform to the dimensions specified in the applicable drawings (see 5.2).

2.3 Performance.

2.3.1 Durometer hardness. The durometer hardness of the boot shall be as specified in ASTM D2000 M4AA 510 A13B13 or ASTM D2000 M4AA 610 A13B13 for the applicable rubber compound. Boots shall be tested for durometer hardness in accordance with ASTM D2240.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 2530

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2.3.2 Volume change. When tested for volume change the volume of test specimen shall neither increase by more than 20 percent (%) nor decrease by more than 5%. After testing, the durometer hardness shall not increase by more than 5 points nor decrease by more than 10 points. Boots shall be tested for volume change by immersion in liquids conforming to SAE J1703. The test fluid temperature shall be 121 ± 3 degrees Celsius (°C). The time of immersion shall be 70 ± 1/4 hours. Specimen shall be approximately 6.45 square centimeters in area. The change in volume of the specimen due to immersion shall be calculated as follows:

$$\text{Change in volume, percent,} = \frac{(W_3 - W_4) - (W_1 - W_2)}{(W_1 - W_3)} \times 100$$

Where:

W_1 is the weight of the specimen in air before immersion, grams.
W_2 is the weight of the specimen in distilled water before immersion, grams.
W_3 is the weight of the specimen in air after immersion, grams.
W_4 is the weight of the specimen in distilled water after immersion, grams.

2.3.3 Heat stroking. Boots shall show no evidence of flex cracks, extending through the wall thickness and fit around cylinder and push rod, after being tested for heat stroking in an oven of the type described in ASTM D573. To conduct the test, two boots shall be mounted on the stroking device. The push rods shall then be cycled at the rate of 1000 ± 100 strokes per hour for 120,000 cycles. Stroke length shall be 80% of the maximum stroke for master cylinders and 5 millimeters for wheel cylinders. The ambient air temperature shall be 70 ± 1°C. No oil shall contact the boots during stroking.

2.3.4 Low temperature flexibility. Boots shall show no evidence of damage, or cracks after being tested for low temperature flexibility. To conduct this test, two boots shall be mounted on a fixture, placed in a cold chamber, and held at a temperature of minus 18.3 ± 2°C for not less than 70 hours. The boots shall be so mounted that relative motion between each boot and push rod will not occur. The push rods shall then be actuated through the maximum stroke, for which the cylinder was assigned, for a total of 10 cycles.

2.3.5 Heat resistance. After oven aging, the durometer hardness of the boot shall not be greater than 15 points higher or not more than 5 points lower than the initial durometer hardness. The rubber, after cooling, shall not be tacky and there shall be no cracking when folded. Two boots shall be tested. The test shall consist of suspending one of the two boots in the oven for 70 ± 1/4 hours at 100 ± 3°C and suspending the other boot in the oven for 24 ± 1/4 hour at 121 ± 3°C. A circulating oven described in ASTM D573 shall be provided.

2.3.6 Ozone resistance. Boots shall exhibit no evidence of cracking or rupture under seven power magnification after being exposed to ozone concentration test. Two boots shall be assembled on a hydraulic cylinder or equivalent which will provide a 12 to 15% stretch in both the boot-head and push-rod mounting areas. The boots shall then be conditioned in an ozone-free atmosphere at room temperature for not more than 45 minutes. Samples shall

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then be placed in an ozone chamber with the ozone concentration maintained at 50 ± 5 parts per 100,000,000 parts of air by volume, at a temperature of 38 ± 1°C, for a period of 168 ± 1/2 hours. The boots shall be examined daily for cracking and rupture under seven power magnification.

2.4 Identification and marking. Unless otherwise specified (see 5.2), identification and marking shall be legible and permanent and shall include as a minimum the manufacturer's CAGE code, the part number, and the symbol "OZ".

2.5 Workmanship. Dust boots shall be free of blisters, cracks and embedded foreign material.

3.0 Quality assurance provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Non-Government publications. ASTM D573 "Standard Test Method for Rubber-Deterioration in an Air Oven"; ASTM D2000 "Standard Classification System for Rubber Products in Automotive Applications"; ASTM D2240-91 "Standard Test Method for Rubber Property-Durometer Hardness" are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103; SAE J1703-91 "Motor Vehicle Brake Fluid, Standards"; are available from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this commercial item description.
b. Issue of the Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 5.1).
c. Title, number, and revision letter of the applicable engineering drawing (see 2.2).
d. Identification marking required (see 2.4).
e. The part number (see 2.4).
f. Selection of applicable levels of preservation, packaging, packing, labeling and marking (see 4).

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5.3 Cross-reference. Boots conforming to this CID are interchangeable/ substitutable with boots conforming to MIL-B-45326C(AT).

5.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to maximum extent practicable.

Custodian:
Army - AT

Review activities:
Army - MI
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Preparing activity:
Army - AT

(Project 2530-0345)

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